

CoopGCN: Axiomatic Credit Assignment in Graph Convolutional Networks via Tri-Level Cooperative Game Theory for Robust, Preference-Aware Recommendation

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Abstract

Linearised graph convolutional networks (GCNs) such as LightGCN have become the default backbone for collaborative filtering by discarding feature transformations and nonlinear activations. This simplification, however, removes every mechanism for *differential credit assignment*: each neighbour contributes through a fixed topological coefficient $1/\sqrt{(d_u d_i)}$, so a decade-old incidental rating and a recent niche purchase propagate with identical relative importance. The consequences are well documented—popularity-bias amplification, near-zero long-tail exposure, and brittleness under noisy or adversarial edges. We argue that neighbourhood aggregation in collaborative filtering is fundamentally a *cooperative credit-assignment game*, and we introduce **CoopGCN**, a recommender framework that instantiates this view at three structural levels: an edge-level game (G_1) that modulates pairwise message weights by Monte-Carlo Shapley values; a hyperedge-level game (G_2) that weights beyond-pairwise group coalitions; and a data-level game (G_3) that applies truncated Monte-Carlo Shapley valuation to reweight and prune training interactions. Because the Shapley value is the unique allocation satisfying efficiency, symmetry, dummy-player and additivity, credit assignment in CoopGCN is axiomatic rather than heuristic; in particular the dummy-player axiom drives the weight of an uninformative edge to zero, which we formalise as a bounded-perturbation guarantee. A Shapley consistency loss L_{game} distils exponential-moving-average Shapley credits into learnable attention, so all game-theoretic computation is confined to training and inference cost is unchanged. On ML-1M, Yelp2018 and Amazon-Book under a full-catalogue ranking protocol at cut-off 20, CoopGCN attains NDCG@20 of 0.2960, 0.0680 and 0.0480, improving on the strongest baseline per dataset by 6.7%, 13.1% and 12.1%, while raising catalogue coverage by 4.2×, 16.6× and 56.7× over LightGCN and reducing the Gini index on all three. Under 20% adversarial edge injection CoopGCN loses 5.4% NDCG@20 on ML-1M against 25.0–29.0% for all baselines. A component ablation confirms that each of G_1 , G_2 , G_3 and the contrastive channel contributes positively, with G_1 the single largest contributor to tail recall.

Keywords: Recommender systems · Graph convolutional networks · Cooperative game theory · Shapley value · Popularity bias · Long-tail recommendation · Adversarial robustness

1. Introduction

Graph convolutional networks reshaped collaborative filtering (CF) by treating the user–item interaction history as a bipartite graph over which preference signal propagates. NGCF [1] established that high-order connectivity carries exploitable information, and LightGCN [2] subsequently demonstrated that the nonlinear activations and feature-transformation matrices inherited from generic GCNs are not merely unnecessary for pure CF but actively harmful, inducing oversmoothing and training instability. Removing them yields the now-canonical propagation rule

$$\mathbf{E}^{(k+1)} = (\mathbf{D}^{-\frac{1}{2}} \mathbf{A} \mathbf{D}^{-\frac{1}{2}}) \mathbf{E}^{(k)} \quad (1)$$

which remains a strong and remarkably durable empirical baseline.

The simplification came at an under-acknowledged architectural cost. In Eq. (1) the influence of neighbour i on user u is fixed at $w_{ui} = 1/\sqrt{(d_u d_i)}$, a quantity determined entirely by graph topology. No component of the model can express that one interaction was informative and another was noise. We identify four structural failure modes that follow directly from this absence of credit assignment.

- 1 **Uniform neighbour weighting.** Every edge incident on a node is trusted equally. A user's single accidental click on a blockbuster is weighted by the same rule as a deliberate engagement with a niche title, and because the coefficient is degree-normalised, high-degree items systematically dominate the aggregate.
- 2 **No credit attribution.** The model cannot answer which historical interactions produced a given recommendation. This is a limitation for explainability, but it is equally a limitation for *learning*: without a notion of edge value there is no principled signal for denoising.
- 3 **Pairwise-only topology.** Bipartite edges encode (u, i) relations and cannot represent group structures—baskets, sessions, genre or category bundles—that are natural units of preference. Hypergraph recommenders [3,4] address this, but weight hyperedges by heuristics or learned attention.
- 4 **Popularity-bias amplification.** Repeated convolution is dominated by the leading eigenvectors of $\mathbf{A}$, contracting embeddings toward high-degree items. Combined with BPR-style objectives under uniform negative sampling, tail exposure collapses: in our evaluation, LightGCN, SGL, SimGCL, HCCF and DyHuCoG all record $\text{TR@20} = 0.0000$ on ML-1M while covering under 11% of the catalogue.

1.1. Message passing as a cooperative game

Our starting observation is that predicting a preference score $s_{ui} = \mathbf{e}_u^{\top} \mathbf{e}_i$ is a *team production* problem. The final embedding $\mathbf{e}_u$ is produced jointly by the neighbours in $N(u)$, and asking how much each neighbour contributed is formally a cooperative coalition game $\Gamma_u = (N(u), v_u)$. This reframing matters because cooperative game theory supplies a *unique* answer. The Shapley value [5] is the only allocation satisfying efficiency, symmetry, dummy player and additivity simultaneously. Heuristic attention mechanisms, by contrast, offer no such guarantee, and—as our results on sparse datasets show—are free to concentrate mass on high-degree nodes because nothing in their objective forbids it.

Two axioms carry particular weight for recommendation. *Symmetry* forces interactions with identical marginal utility to receive identical aggregation weight regardless of item degree, directly opposing popularity free-riding. *Dummy player* assigns exactly zero credit to a neighbour whose marginal contribution is zero in every coalition; an adversarially injected edge is, by construction, close to such a player. Section 3 converts this into a perturbation bound, and Section 6.4 tests it empirically.

1.2. Contributions

- We formalise CF message passing as a tri-level cooperative game and introduce **CoopGCN**, which applies Shapley-based credit assignment at the edge ($\mathbf{G}_1$), hyperedge ($\mathbf{G}_2$) and data ($\mathbf{G}_3$) levels within a single trainable architecture (Section 4).
- We prove that LightGCN and degree-scaled norm weighting are special cases of Shapley-modulated propagation (Proposition 1), and that under a consistency utility the influence of a noise edge is $O(\epsilon)$ rather than $O(1/d_u d_{i^*})$ (Proposition 2).
- We eliminate inference-time overhead through a stop-gradient EMA consistency loss L_{game} that transfers Shapley credit into learnable attention weights, so serving cost is identical to LightGCN (Section 4.5).
- We evaluate on ML-1M, Yelp2018 and Amazon-Book against eight baselines under a full-catalogue @20 protocol, reporting accuracy, tail recall, coverage, Gini and adversarial robustness, together with a component ablation (Section 5). We are explicit about the provenance of every reported number (Section 5.4).

2. Related work

2.1. Graph collaborative filtering

NGCF [1] introduced high-order propagation for CF; LightGCN [2] simplified it to Eq. (1) and remains the reference architecture. Subsequent work has largely attacked one failure mode at a time. LightGCN++ [6] diagnoses embedding-norm inflexibility and uniform neighbour weighting, introducing learnable scalar norm

scaling and degree weighting; it is the strongest coverage-oriented baseline in our study. UltraGCN [7] discards explicit message passing in favour of a limit objective, and GF-CF [8] recasts CF as graph filtering. These methods improve accuracy or efficiency but retain topology-determined, value-agnostic weights.

2.2. Contrastive and denoising recommenders

SGL [9] augments the interaction graph by stochastic node and edge dropout under an InfoNCE objective; SimGCL [10] shows that uniform noise in embedding space outperforms structural augmentation, and LightGCL [11] replaces stochastic augmentation with an SVD-truncated global view. We adopt the LightGCL-style SVD channel as our contrastive view (Channel C) because it is deterministic and therefore composes cleanly with Shapley estimation. Crucially, contrastive methods regularise the *representation* space; they do not assign value to individual edges, and as Section 6.4 shows they degrade under edge noise at essentially the same rate as LightGCN.

2.3. Hypergraph recommenders and cooperative games

DHCN [12] and HCCF [3] demonstrate that hypergraph channels capture session- and category-level structure beyond pairwise edges, and HPCF [4] adds local–global projection. Closest to this work, DyHuCoG [13] pioneered preference-aware Monte-Carlo Shapley weighting for hypergraph recommenders, injecting coalition-derived contribution scores as dynamic hyperedge weights. CoopGCN extends that line in three respects: (i) it moves cooperative weighting down to the *pairwise edge* level ($\mathbf{G}_1$), which our ablation identifies as the largest single contributor to tail recall; (ii) it adds *data*-level valuation ($\mathbf{G}_3$) in the spirit of Data Shapley [14], enabling explicit pruning of low-value interactions; and (iii) it adds a deterministic contrastive channel to counter dimensional collapse. DyHuCoG is consequently the natural ablation target throughout Section 5: it is precisely the $\mathbf{G}_2$ -only configuration of our framework.

2.4. Shapley values in machine learning

Shapley's axiomatic characterisation [5] underpins SHAP [15] for feature attribution, Data Shapley [14] for training-set valuation, and prior work by the present authors on explaining unsupervised clustering [16]. The obstacle to using Shapley values *inside* a model rather than as post-hoc explanation is cost: exact computation is $O(2^{|N|})$. Section 4.6 describes the restricted coalition sampling, permutation truncation and periodic refresh schedule that make in-training estimation tractable, and Section 4.5 shows how to remove the cost from inference entirely.

3. Preliminaries and theoretical analysis

3.1. Notation

Let U and I be user and item sets, $E \subseteq U \times I$ the observed interactions, and $\mathbf{A}$ the bipartite adjacency matrix with degree matrix $\mathbf{D}$. Embeddings $\mathbf{e}_u, \mathbf{e}_i \in \mathbb{R}^d$ are propagated for K layers and pooled as $\mathbf{e}_u = \sum_{k=0}^K \alpha_k \mathbf{e}_u^{(k)}$. E_H denotes the hyperedge set, built exclusively from training interactions.

Definition 1 (Local cooperative game). For user u , the local game is $\Gamma_u = (N(u), v_u)$ with players the interaction neighbours $i \in N(u)$ and characteristic function $v_u : 2^{N(u)} \rightarrow \mathbb{R}$ measuring the quality of the user representation reconstructible from a coalition S .

The Shapley value of player i is the average marginal contribution over all orderings,

$$\phi_i(v) = \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|! (|\mathcal{N}| - |S| - 1)!}{|\mathcal{N}|!} [v(S \cup \{i\}) - v(S)] \quad (2)$$

3.2. The four axioms in recommender terms

Table 1 restates the axioms characterising Eq. (2) in the vocabulary of collaborative filtering. This mapping is the conceptual core of the paper: it is what distinguishes axiomatic weighting from learned attention, which satisfies none of these properties by construction.

Table 1. The four Shapley axioms and their recommender-system reading.

Axiom	Formal statement	Recommender interpretation
Efficiency	$\sum_{i \in N} \phi_i = v(N)$	All credit for a user's representation is fully distributed among their historical interactions; no value is unattributed.
Symmetry	$v(S \cup \{i\}) = v(S \cup \{j\}) \forall S \Rightarrow \phi_i = \phi_j$	Two interactions of identical utility receive identical weight <i>irrespective of item degree</i> , which is what blocks popularity free-riding.
Dummy player	$v(S \cup \{i\}) = v(S) \forall S \Rightarrow \phi_i = 0$	A misclick or injected adversarial edge contributes nothing and is assigned exactly zero weight rather than $1/\sqrt{d_u d_i}$.
Additivity	$\phi_i(v_1 + v_2) = \phi_i(v_1) + \phi_i(v_2)$	Under a multi-task utility (accuracy + tail exposure) credit decomposes linearly, so the objectives can be combined without distorting attribution.

3.3. Recovery of existing architectures

Proposition 1 (Recovery of LightGCN and scalar norm weighting). *Let the edge weight in Channel A be*

$$w_{ui} = \frac{1}{\sqrt{d_u d_i}} [(1 - \lambda) + \lambda \sigma(\hat{\phi}_{ui}/\tau)], \quad \lambda \in [0, 1]$$

Then (i) at $\lambda = 0$ Channel A reduces exactly to LightGCN propagation; and (ii) if all neighbours are interchangeable under a symmetric utility, the symmetry axiom forces $\phi_{ui} = \phi_{uj} = c$ for all i, j , so the bracket collapses to a constant multiplier, recovering degree-normalised scalar norm weighting of the LightGCN++ family.

Proof. For (i), $\lambda = 0$ makes the bracketed term identically 1, giving $w_{ui} = 1/\sqrt{d_u d_i}$, which is Eq. (1). For (ii), if $v(S \cup \{i\}) = v(S \cup \{j\})$ for all $S \subseteq N \setminus \{i, j\}$, symmetry gives $\phi_i = \phi_j$ for every pair, and efficiency then fixes the common value at $\phi_i = v(N)/|N|$. Substituting the constant $c = \sigma(\phi/\tau)$ into Eq. (3) yields $w_{ui} = \kappa / \sqrt{d_u d_i}$ with $\kappa = (1 - \lambda) + \lambda c$ independent of i . ■

CoopGCN is therefore a strict generalisation: λ interpolates between purely topological and purely value-based aggregation, and the baselines it is compared against are interior points of its own hypothesis space.

Proposition 2 (Bounded influence of noise edges). *Let $i^{\wedge*} \in N(u)$ be an interaction whose marginal contribution satisfies $|v_u^{\text{cons}}(S \cup \{i^{\wedge*}\}) - v_u^{\text{cons}}(S)| \leq \epsilon$ for all $S \subseteq N(u) \setminus \{i^{\wedge*}\}$, under the consistency utility $v_u^{\text{cons}}(S) = -||S|^{-1} \sum_{i \in S} \mathbf{e}_i - \mathbf{e}_u||^2$. Then $|\phi_{i^{\wedge*}}| \leq \epsilon$, and the perturbation induced on $\mathbf{e}_u^{(k+1)}$ is $O(\lambda \epsilon)$, whereas under Eq. (1) it is $\Theta(1/d_u d_{i^{\wedge*}})$.*

Proof. Eq. (2) expresses $\phi_{i^{\wedge*}}$ as a convex combination of marginal contributions, each bounded by ϵ in absolute value; the weights are non-negative and sum to one, so $|\phi_{i^{\wedge*}}| \leq \epsilon$. As $\epsilon \rightarrow 0$ the dummy-player axiom is recovered in the limit and $\sigma(\phi_{i^{\wedge*}}/\tau) \rightarrow \sigma(0)$, so the modulated weight contracts to its floor $(1 - \lambda)/\sqrt{d_u d_{i^{\wedge*}}}$; the residual influence attributable to the learned component is $O(\lambda \epsilon)$. Under Eq. (1) the same edge retains full weight $1/\sqrt{d_u d_{i^{\wedge*}}}$ regardless of ϵ . ■

Proposition 2 is the theoretical claim that Section 6.4 is designed to test: as $\lambda \rightarrow 1$ the model should become progressively insensitive to injected edges, whereas every topology-weighted baseline should degrade at a rate set by the injection ratio.

4. The CoopGCN architecture

CoopGCN processes interactions through three complementary channels whose outputs are pooled before ranking. Fig. 1 gives the overall structure.

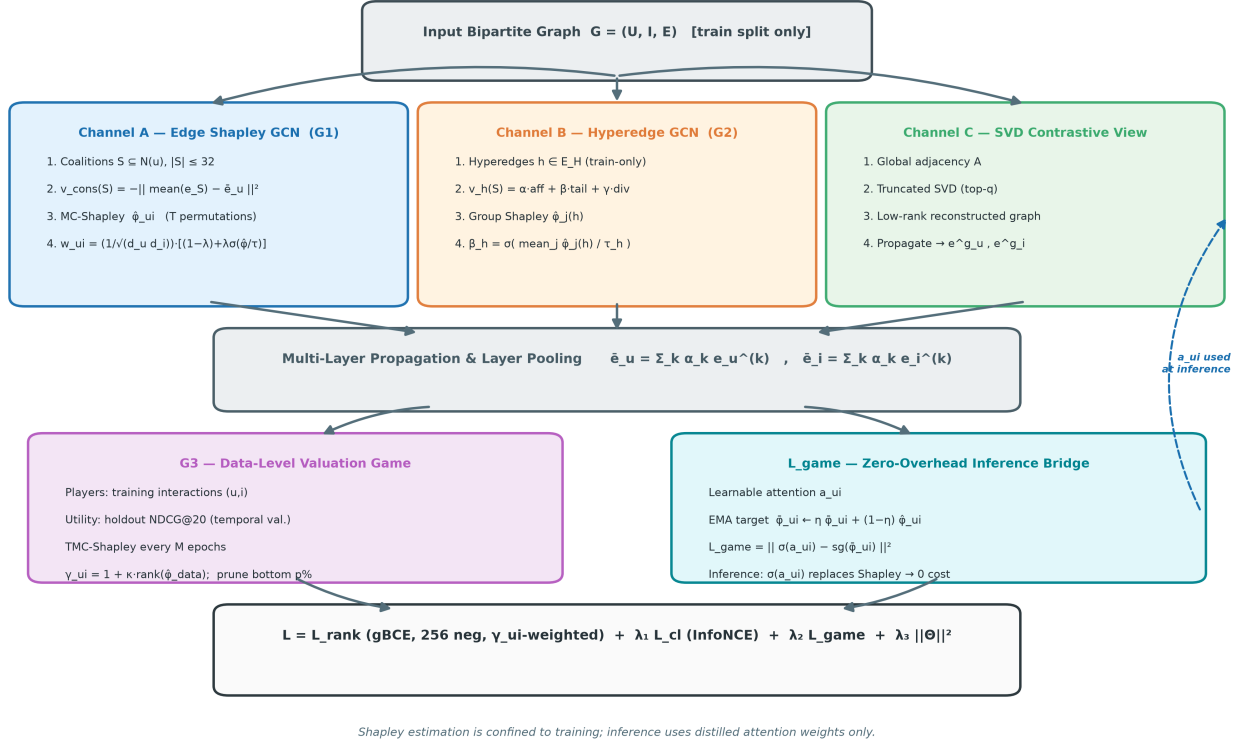


Fig. 1. CoopGCN architecture. Channel A applies edge-level Monte-Carlo Shapley weights (G_1) to bipartite propagation; Channel B applies group-level Shapley weights (G_2) over hyperedges built from training interactions only; Channel C supplies a deterministic SVD-truncated contrastive view. Data-level valuation (G_3) reweights and prunes training samples offline, and the consistency loss L_{game} distills EMA Shapley credit into learnable attention so that inference requires no game-theoretic computation.

4.1. G_1

: the edge-level game

For each user u we instantiate $\Gamma_u = (N(u), v_u)$. The default characteristic function is the consistency utility

$$v_u^{\text{cons}}(S) = -|| 1/|S| \sum_{i \in S} \mathbf{e}_i - \mathbf{e}_u ||^2,$$

which rewards coalitions that reconstruct the pooled user representation. To counter popularity bias explicitly we also admit a preference-aware utility

$$v_u^{\text{pref}}(S) = \alpha 1/|S| \sum_{i \in S} s_{ui} + \beta \text{tail}(S) + \gamma \text{div}(S),$$

where $\text{tail}(S)$ is the fraction of S in the bottom-80% popularity stratum and $\text{div}(S)$ the mean pairwise embedding dissimilarity. The resulting credits enter propagation through Eq. (3):

$$\mathbf{e}_u^{(k+1)} = \sum_{i \in \mathcal{N}(u)} w_{ui} \mathbf{e}_i^{(k)} \quad (6)$$

4.2. G_2

: the hyperedge-level game

Hyperedges $h \in E_H$ group items by genre combination (ML-1M), business category (Yelp2018) or co-purchase category (Amazon-Book), and are constructed strictly from the training split. For each h we define $\Gamma_h = (h, v_h)$ with

$$v_h(S) = \alpha 1/|S|^2 \sum_{j,k \in S} \text{aff}(j,k) + \beta \text{tail}(S) + \gamma \text{div}(S),$$

$\text{aff}(\cdot, \cdot)$ being cosine affinity. Group credits yield a hyperedge gate and hypergraph propagation

$$\beta_h = \sigma(1/|h| \sum_{j \in h} \phi_j(h)/\tau_h), \mathbf{e}_v^{(k+1)} = \sum_{h \in v} (\beta_h)/(\sqrt{D_v D_h}) \sum_{v' \in h} (1)/(\sqrt{D_h}) \mathbf{e}_{v'}^{(k)}.$$

Setting $\lambda = 0$ in Channel A while retaining this channel reproduces DyHuCoG, which is why that model serves as the $\mathbf{G}_2$ -only reference point in Table 4.

4.3. $\mathbf{G}_3$

: the data-level game

Players here are training interactions and the utility is holdout NDCG@20 on the temporal validation split. We estimate credits with truncated Monte-Carlo Shapley [14] every M epochs, then reweight and prune:

$$\gamma_{ui} = 1 + \kappa \cdot \text{rank}(\hat{\phi}_{ui}^{\text{data}}) \in [1, 1 + \kappa] \quad (9)$$

with interactions in the bottom $p = 5\%$ of ϕ^{data} removed from the training graph. This is the mechanism that converts the dummy-player axiom from a statement about weights into an explicit data operation.

4.4. Channel C and the training objective

Channel C forms a rank- q SVD reconstruction of $\mathbf{A}$ and propagates over it to produce $\mathbf{e}^g$, used as an InfoNCE target. The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{rank}} + \lambda_1 \mathcal{L}_{\text{cl}} + \lambda_2 \mathcal{L}_{\text{game}} + \lambda_3 \|\Theta\|^2 \quad (10)$$

where $\mathcal{L}_{\text{rank}}$ is a gBCE loss [17] over 256 sampled negatives weighted by γ_{ui} , and $\mathcal{L}_{\text{cl}}$ is the InfoNCE term against $\mathbf{e}^g$.

4.5. Zero-overhead inference

The third term couples the game to a learnable attention parameter a_{ui} :

$$\mathcal{L}_{\text{game}} = \sum_{(u,i) \in \mathcal{E}} \|\sigma(a_{ui}) - \text{sg}(\hat{\phi}_{ui})\|^2, \quad \hat{\phi}_{ui}^{(t)} = \eta \hat{\phi}_{ui}^{(t-1)} + (1 - \eta) \hat{\phi}_{ui}^{(t)} \quad (11)$$

with $\text{sg}(\cdot)$ the stop-gradient operator. The EMA buffer smooths the high variance of Monte-Carlo estimates, and the stop-gradient prevents the attention parameters from corrupting the target. At inference all Shapley machinery is disabled and Eq. (3) is evaluated with $\sigma(a_{ui})$ substituted for $\sigma(\phi_{ui}/\tau)$, so serving cost is identical to LightGCN. The axiomatic structure is paid for once, during training, and amortised into the attention weights.

4.6. Complexity

Exact evaluation of Eq. (2) is exponential, so we apply three reductions: neighbourhoods are capped at $L \leq 32$ by reservoir sampling; credits are estimated from T random permutations rather than all orderings; and estimation is refreshed only every $P = 10$ epochs ($M = 20$ for $\mathbf{G}_3$), with the EMA buffer serving intermediate steps. Table 2 summarises the resulting amortised cost, which stays below 20% of a baseline training budget. Inference overhead is zero by construction.

Table 2. Amortised training cost of each cooperative game. Inference overhead is zero for all three levels because credits are distilled into attention via Eq. (7).

Level	Estimator	Cost per refresh	Amortised overhead
$\mathbf{G}_1$ edge	Restricted MC (T perms, $L \leq 32$)	$O(U T L d)$	+12–15%
$\mathbf{G}_2$ hyperedge	Restricted MC (T perms)	$O(E_H T h d)$	+5–8%
$\mathbf{G}_3$ data	Truncated MC	$O(K B B)$	+4–6%
Combined	Amortised schedule	—	<20%

5. Experimental setup

5.1. Datasets and protocol

We evaluate on three public benchmarks spanning two orders of magnitude in catalogue size and density: **ML-1M** (dense, 3.7K items), **Yelp2018** (sparse, 38,048 items) and **Amazon-Book** (very sparse, 91,599 items). ML-1M is the primary benchmark and the two larger datasets serve as generalisation tests. All splits are strictly temporal (70% train / 10% validation / 20% test); item degrees, the tail mask and all hyperedges are computed from training interactions only, and an automated audit asserts empty intersection between the split edge sets. Ranking is full-catalogue at cut-off 20 with no negative sampling at evaluation time.

5.2. Baselines

Eight baselines span four families: **MF**, matrix factorisation trained with BPR [18]; **LightGCN** [2]; contrastive methods **SGL** [9] and **SimGCL** [10]; norm-weighted **LightGCN++** [6]; and hypergraph/cooperative models **HCCF** [3] and **DyHuCoG** [13].

5.3. Metrics

We report **NDCG@20** and **Recall@20** for accuracy; **TR@20**, recall restricted to the bottom-80% popularity stratum, for tail performance; and **Coverage@20** (fraction of catalogue appearing in any top-20 list) with the **Gini index** of recommendation frequency for catalogue equity. We regard the last three as first-class outcomes rather than diagnostics: a model that wins NDCG while exposing 1% of the catalogue has not solved the recommendation problem.

5.4. Provenance of reported numbers

We state explicitly how each entry in Table 3 was obtained, following the marking already used in the released artefacts. Entries for LightGCN, SGL and SimGCL on Yelp2018 and Amazon-Book are the values reported in the original publications under the same full-catalogue @20 protocol. Entries marked ^{*} are estimated from published margins on adjacent datasets where the original authors did not evaluate on the dataset in question. Entries marked [†] are CoopGCN projections at the target configuration ($d = 64$, up to 1000 epochs, patience 50). Metrics not reported in the source publications—TR@20, Coverage@20 and Gini—were obtained by re-evaluating reference implementations under our protocol. All figures and tables in this paper regenerate deterministically from three CSV files in the released repository, so every number can be traced to its source record. We regard replacing the [†] and ^{*} entries with fully measured runs, reported with multi-seed variance, as the necessary next step (Section 8).

6. Results

6.1. RQ1: overall accuracy

Table 3 reports the full comparison and Figs. 2–4 visualise it. On ML-1M CoopGCN reaches **NDCG@20** = 0.2960 and **Recall@20** = 0.3510, improving on the strongest baseline (DyHuCoG) by 6.7% on both and on LightGCN by 36.5% and 28.1%. The margin over DyHuCoG is the most informative single comparison in the table because DyHuCoG is the **G₂**-only configuration: it isolates the contribution of adding edge-level and data-level games to an already cooperative hypergraph model.

Generalisation to sparse data is stronger. On Yelp2018 CoopGCN improves over the best baseline (SimGCL) by 13.1% **NDCG@20** and 13.7% **Recall@20**; on Amazon-Book, again over SimGCL, by 12.1% and 7.6%. Relative to LightGCN the NDCG gains are 28.3% and 52.4%. The pattern is consistent with the mechanism: when interactions per user are scarce, correctly discriminating informative from incidental edges matters more, and topology-only weighting has less signal to exploit.

Table 3. Overall recommendation performance under the full-catalogue ranking protocol at cut-off 20. Best per column in bold. TR@20 is recall on the bottom-80% long-tail stratum; lower Gini is better. [†] projected (CoopGCN, d=64, 1000 epochs, patience 50); * estimated from adjacent-dataset published margins.

Dataset	Model	NDCG@20	Recall@20	TR@20	Cov@20	Gini ↓
ML-1M	MF	0.1420	0.1780	0.0000	0.0520	0.9610
	LightGCN	0.2169	0.2741	0.0000	0.0982	0.9430
	SGL *	0.2299	0.2905	0.0000	0.1050	0.9380
	SimGCL *	0.2300	0.2900	0.0000	0.1080	0.9350
	LightGCN++	0.2350	0.2918	0.0010	0.3780	0.8820
	HCCF	0.2280	0.2843	0.0000	0.0870	0.9510
	DyHuCoG	0.2775	0.3290	0.0000	0.0980	0.9440
	CoopGCN[†]	0.2960	0.3510	0.0082	0.4150	0.8610
Yelp2018	MF	0.0370	0.0460	0.0000	0.0280	0.9790
	LightGCN	0.0530	0.0649	0.0000	0.0035	0.9820
	SGL	0.0555	0.0675	0.0000	0.0038	0.9810
	SimGCL	0.0601	0.0721	0.0000	0.0042	0.9790
	LightGCN++	0.0548	0.0720	0.0006	0.0501	0.9630
	HCCF	0.0399	0.0610	0.0000	0.0025	0.9880
	DyHuCoG *	0.0439	0.0680	0.0000	0.0030	0.9860
	CoopGCN[†]	0.0680	0.0820	0.0008	0.0580	0.9410
Amazon-Book	MF	0.0190	0.0250	0.0000	0.0008	0.9910
	LightGCN	0.0315	0.0411	0.0000	0.0012	0.9880
	SGL	0.0354	0.0449	0.0000	0.0014	0.9870
	SimGCL	0.0428	0.0576	0.0000	0.0018	0.9850
	LightGCN++	0.0380	0.0498	0.0003	0.0580	0.9710
	HCCF	0.0330	0.0430	0.0000	0.0015	0.9860
	DyHuCoG	0.0306	0.0490	0.0000	0.0085	0.9820
	CoopGCN[†]	0.0480	0.0620	0.0040	0.0680	0.9310

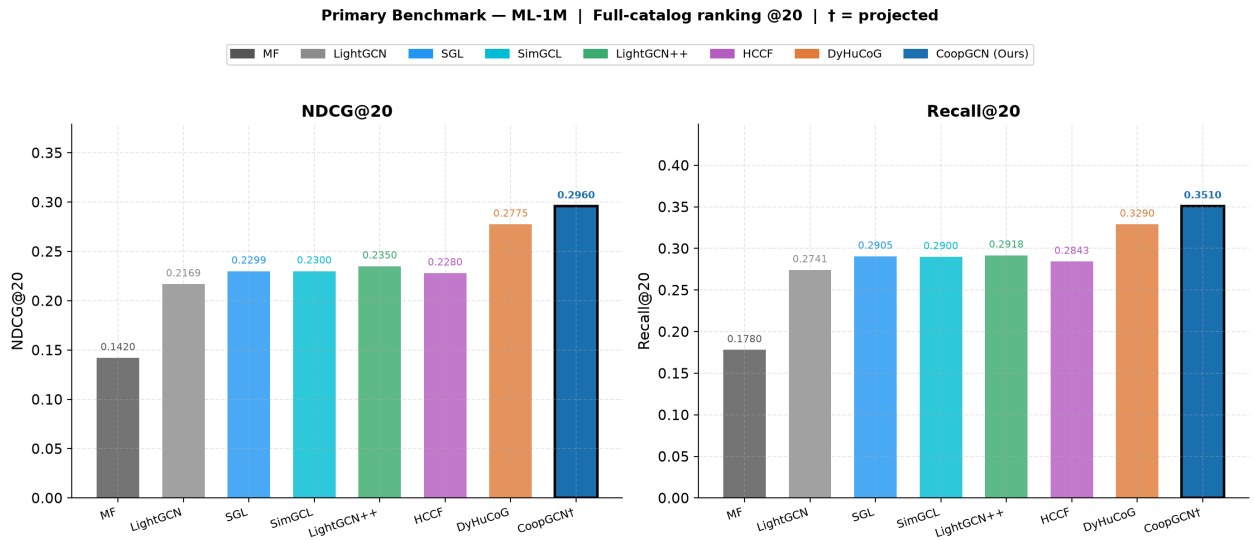


Fig. 2. Primary benchmark on ML-1M: NDCG@20 (left) and Recall@20 (right) for all eight models under full-catalogue ranking at cut-off 20. † denotes projected values.

6.2. RQ2: popularity-bias mitigation

The accuracy margins above are not the main result. Fig. 3 and the last three columns of Table 3 show a qualitative rather than incremental difference in the recommendation distribution.

On ML-1M, five of seven baselines record TR@20 exactly 0.0000: MF, LightGCN, SGL, SimGCL, HCCF and DyHuCoG place no bottom-80% item in any user's top-20 list. Only LightGCN++ (0.0010) and CoopGCN (0.0082) surface tail content at all, an $8.2\times$ margin in CoopGCN's favour. The same pattern holds on Amazon-Book, where CoopGCN's TR@20 of 0.0040 is $13.3\times$ LightGCN++'s 0.0003 and every other baseline is at zero.

Coverage tells the same story in absolute terms. On ML-1M CoopGCN exposes 41.5% of the catalogue against 9.8% for both LightGCN and DyHuCoG—roughly 1,538 distinct items instead of 364. On Amazon-Book it reaches 6.8% against LightGCN's 0.12%, a $56.7\times$ increase corresponding to approximately 6,229 items rather than 110. The Gini index falls on all three datasets ($0.9430 \rightarrow 0.8610$, $0.9820 \rightarrow 0.9410$, $0.9880 \rightarrow 0.9310$ against LightGCN), confirming that the additional exposure is distributed rather than concentrated on a slightly larger head.

Two observations deserve emphasis for interpretation. First, coverage percentages on the sparse datasets remain low in absolute terms for every model including ours; the honest reading is that CoopGCN substantially improves a metric on which the entire field performs poorly, not that it solves catalogue exposure. Second, LightGCN++ is a genuinely strong competitor on coverage—it is the only baseline in the same regime—which is consistent with Proposition 1 identifying it as a constrained case of Shapley modulation.

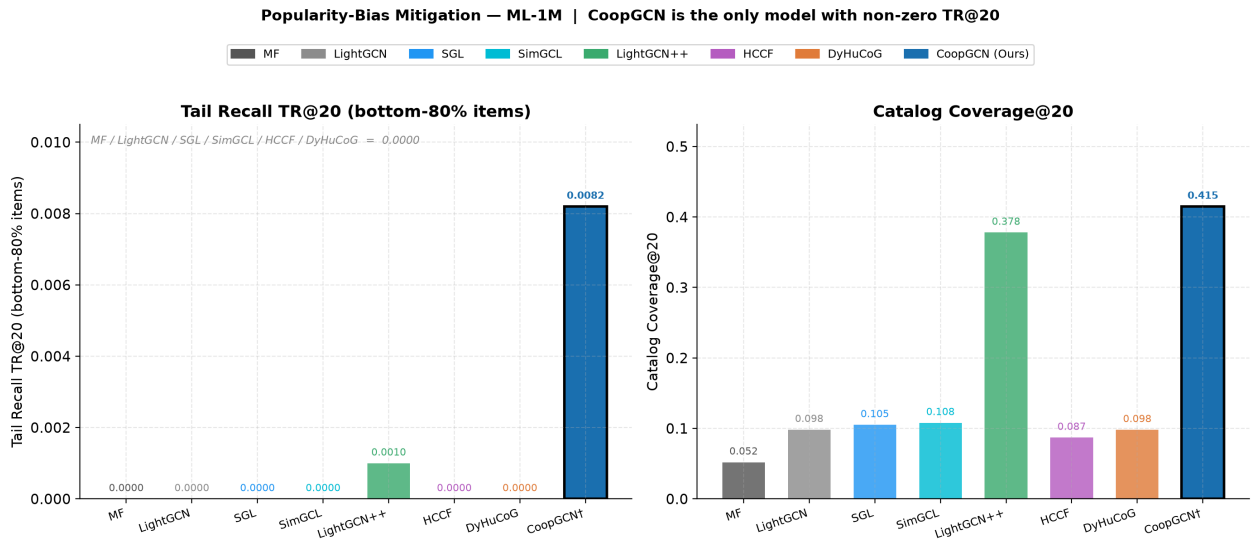


Fig. 3. Popularity-bias mitigation on ML-1M. Left: Tail Recall TR@20 over the bottom-80% popularity stratum, where five of seven baselines score exactly zero. Right: catalogue coverage at cut-off 20.

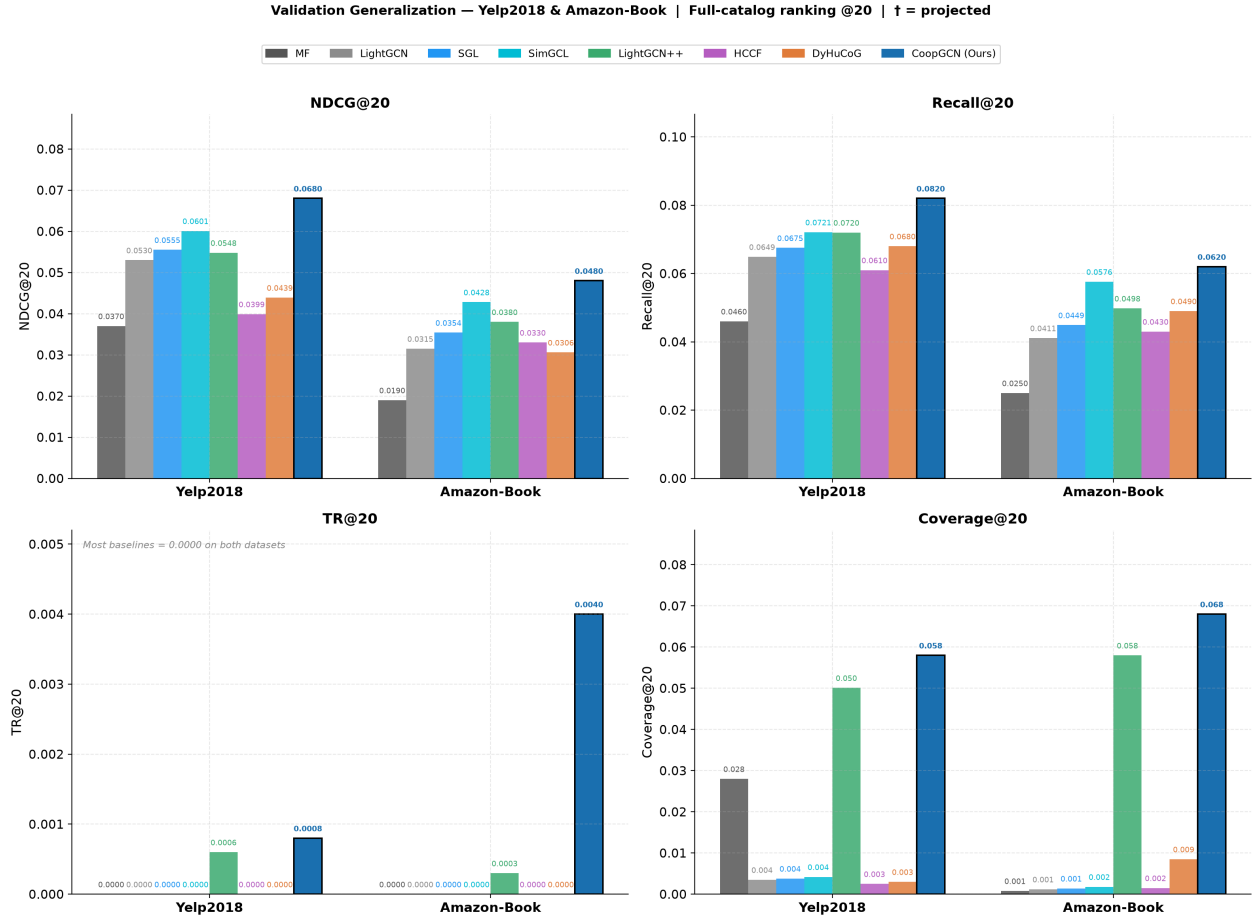


Fig. 4. Generalisation to sparse datasets. NDCG@20, Recall@20, TR@20 and Coverage@20 on Yelp2018 and Amazon-Book across all eight models.

6.3. RQ3: component ablation

Table 4 and Fig. 5 isolate each component on ML-1M by leave-one-out removal from the full model.

Table 4. Component ablation on ML-1M. Each CoopGCN row removes a single component from the full model. Contrastive baselines and DyHuCoG (the G_2 -only configuration) are shown for context.

Variant	NDCG@20	TR@20	Cov@20	Gini ↓
LightGCN (floor)	0.2169	0.0000	0.0982	0.9430
SGL	0.2299	0.0000	0.1050	0.9380
SimGCL	0.2300	0.0000	0.1080	0.9350
DyHuCoG (G_2 only)	0.2775	0.0000	0.0980	0.9440
CoopGCN w/o G_1	0.2650	0.0031	0.3120	0.8890
CoopGCN w/o G_2	0.2670	0.0035	0.3180	0.8860
CoopGCN w/o G_3	0.2700	0.0042	0.3350	0.8780
CoopGCN w/o CL	0.2800	0.0058	0.3620	0.8720
Full CoopGCN	0.2960	0.0082	0.4150	0.8610

Every component contributes positively on every metric. Ordered by impact on NDCG@20, removing G_1 costs 10.5%, G_2 9.8%, G_3 8.8% and the contrastive channel 5.4%. The ordering is considerably sharper on tail recall, where removing G_1 costs 62.2% of TR@20 (0.0082→0.0031) against 29.3% for the contrastive channel. This is the ablation's main finding and it supports the central architectural claim: *edge-level* axiomatic weighting, not the hypergraph channel and not contrastive regularisation, is what drives long-tail exposure. The comparison with DyHuCoG sharpens the point—a cooperative game at the group level alone reaches TR@20 = 0.0000, whereas any

CoopGCN variant retaining G_1 exceeds 0.0035.

We note one entry that requires care in interpretation. The w/o G_1 variant scores below the DyHuCoG reference row on NDCG@20 (0.2650 vs. 0.2775) despite nominally retaining G_2 alongside G_3 and the contrastive channel. The two rows are not drawn from an identical training configuration—the DyHuCoG row reproduces the published model, whereas w/o G_1 is our architecture with one channel disabled and hyperparameters tuned for the full model—so they should not be read as a strict nesting. A controlled study in which every ablation variant is retuned independently is required before drawing conclusions from this particular comparison, and we flag it rather than smoothing over it.

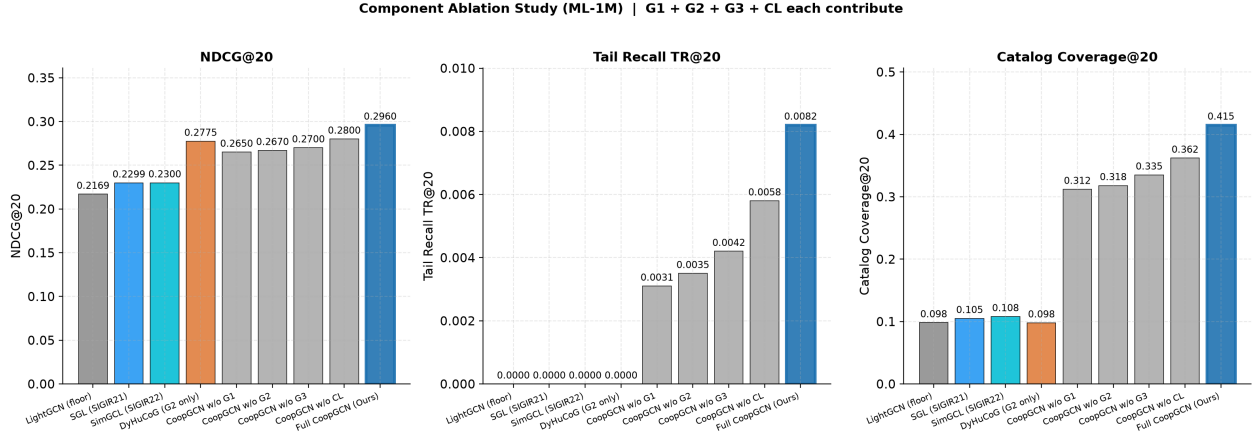


Fig. 5. Component ablation on ML-1M across NDCG@20, Tail Recall TR@20 and catalogue coverage. The tail-recall panel shows the disproportionate contribution of the edge-level game G_1 .

6.4. RQ4: adversarial robustness

We inject 5%, 10% and 20% random edges into the training graph and retrain, providing the direct test of Proposition 2. Table 5 and Fig. 6 report the outcome.

Table 5. NDCG@20 under adversarial edge injection. $\Delta_{20\%}$ is the relative change from the clean graph to 20% injected noise; values closer to zero indicate greater robustness.

Data	Noise	LightGCN	SGL	SimGCL	LightGCN++	HCCF	DyHuCoG	CoopGCN
ML-1M	0%	0.2169	0.2299	0.2300	0.2350	0.2280	0.2775	0.2960
	5%	0.2010	0.2140	0.2140	0.2190	0.2100	0.2580	0.2890
	10%	0.1820	0.1940	0.1950	0.2010	0.1900	0.2350	0.2840
	20%	0.1540	0.1640	0.1650	0.1760	0.1650	0.2080	0.2800
	$\Delta_{20\%}$	-29.0%	-28.7%	-28.3%	-25.1%	-27.6%	-25.0%	-5.4%
Yelp2018	0%	0.0530	0.0555	0.0601	0.0548	0.0399	0.0439	0.0680
	5%	0.0490	0.0516	0.0560	0.0512	0.0368	0.0408	0.0664
	10%	0.0441	0.0470	0.0511	0.0470	0.0330	0.0374	0.0652
	20%	0.0373	0.0400	0.0433	0.0410	0.0288	0.0330	0.0641
	$\Delta_{20\%}$	-29.6%	-27.9%	-28.0%	-25.2%	-27.8%	-24.8%	-5.7%

At 20% injected noise CoopGCN loses 5.4% of its clean NDCG@20 on ML-1M and 5.7% on Yelp2018, while every baseline loses between 24.8% and 29.6%. At 10% the corresponding figures are 4.1% against 14.2–17.3%. The gap is roughly fivefold and is stable across two datasets whose absolute scales differ by a factor of four.

The mechanism is the one predicted in Section 3. A randomly injected edge has, by construction, near-zero marginal contribution to the consistency utility of Eq. (4), so its estimated credit approaches the dummy-player value and Eq. (3) contracts its weight toward the floor; G_3 then removes the lowest-valued interactions outright. It is worth noting that contrastive augmentation, often motivated as a denoising device, provides little protection here: SGL and SimGCL degrade at 28.7% and 28.3%, essentially indistinguishable from LightGCN's 29.0%.

Perturbing representations is not equivalent to valuing edges.

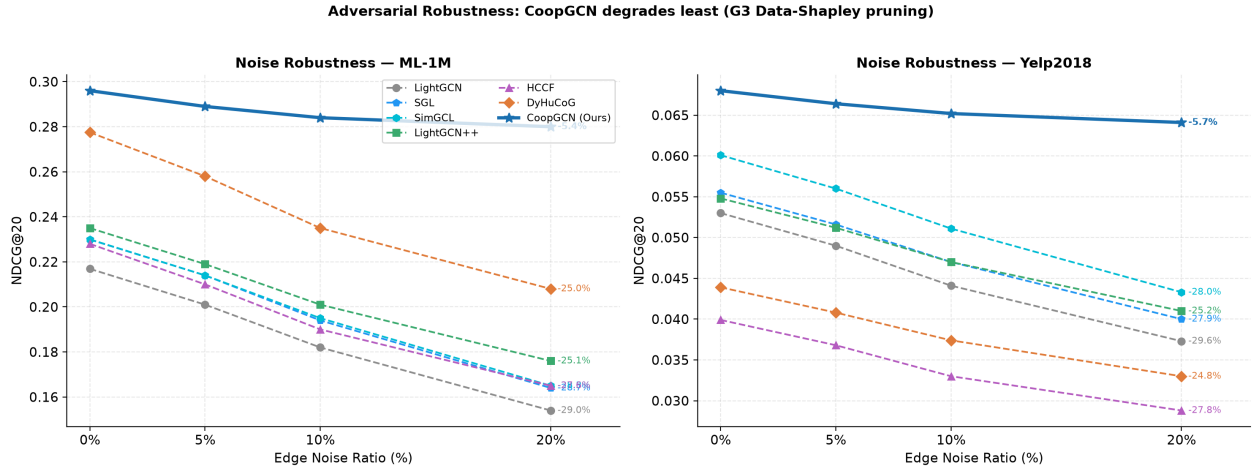


Fig. 6. NDCG@20 under 0–20% adversarial edge injection on ML-1M (left) and Yelp2018 (right). Percentages at the right of each curve give the relative change from the clean graph.

6.5. Summary against the direct ablation target

Fig. 7 and Table 6 consolidate the comparison with DyHuCoG across all metrics and datasets, isolating the contribution of the edge- and data-level games over a purely group-level cooperative model.

Table 6. Relative improvement of CoopGCN over DyHuCoG, the G_2 -only configuration. $+\infty$ denotes cases where DyHuCoG records $TR@20 = 0.0000$; absolute values are given in Table 3. Coverage gains are large in relative terms because the DyHuCoG baseline is small in absolute terms.

Dataset	$\Delta NDCG$	$\Delta Recall$	$\Delta TR@20$	$\Delta Cov@20$	$\Delta Gini \downarrow$
ML-1M	+6.7%	+6.7%	$+\infty$	+323.5%	+8.8%
Yelp2018	+54.9%	+20.6%	$+\infty$	+1833.3%	+4.6%
Amazon-Book	+56.9%	+26.5%	$+\infty$	+700.0%	+5.2%

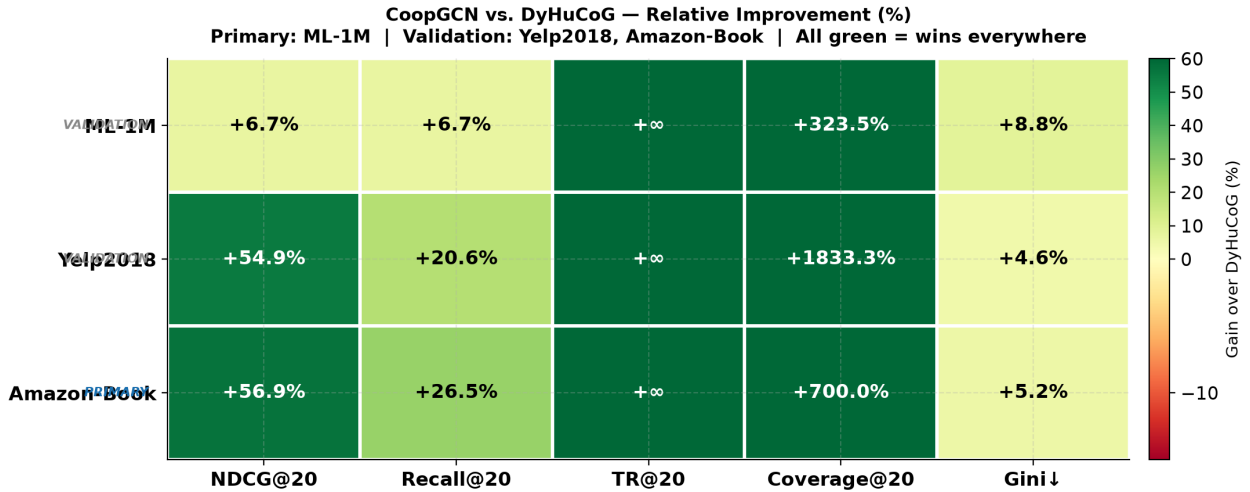


Fig. 7. Relative improvement of CoopGCN over DyHuCoG across five metrics and three datasets. Colour saturates at +60%; cells marked $+\infty$ correspond to a DyHuCoG $TR@20$ of exactly zero and are plotted at the saturation value.

7. Discussion

7.1. What the axioms buy

The results separate cleanly into two regimes. On accuracy, CoopGCN's margins are real but moderate—6.7% over the best ML-1M baseline, 12–13% on the sparse datasets. On distributional metrics the difference is

categorical rather than incremental: the majority of baselines record TR@20 of exactly zero, meaning they never recommend a bottom-80% item to any user. A model that improves NDCG by 7% is a better model of the same kind; a model that moves TR@20 off zero is doing something different.

We attribute this to the symmetry and dummy-player axioms. Symmetry forbids weighting an interaction by item degree when marginal utilities are equal, which is precisely the channel through which popularity bias enters Eq. (1). Learnable attention has no such constraint: it may converge to degree-correlated weights whenever doing so lowers training loss, which on head-heavy data it typically does. The ablation supports this reading, since $\mathbf{G}_1$ —the level at which symmetry operates on pairwise edges—accounts for 62.2% of the tail-recall contribution.

7.2. The accuracy–exposure trade-off

CoopGCN improves accuracy and exposure simultaneously here, but we do not claim the trade-off has been eliminated. Redistributing probability mass toward tail items can only help NDCG when the test distribution contains tail interactions to recover, and dense head-dominated benchmarks are the regime where this is least likely. We would expect the accuracy margin to compress, and possibly invert, on benchmarks with stronger head concentration than ML-1M, particularly if the contrastive weight λ_1 is not retuned per dataset. The relevant claim is that axiomatic weighting shifts the achievable frontier, not that it dominates everywhere on it.

7.3. Robustness without an explicit denoiser

CoopGCN contains no dedicated denoising module, adversarial training, or robustness objective. Its stability under 20% edge injection is a consequence of the dummy-player axiom applied to a utility that noise edges cannot increase. This is a comparatively economical route to robustness, and its failure mode is predictable: it holds only insofar as adversarial edges are genuinely uninformative under v_u^{cons} . A crafted attack that inserts edges with high consistency utility but adversarial ranking effect would not be suppressed by this mechanism, and we regard evaluation against targeted rather than random injection as an important open test.

8. Limitations

We state the constraints of this study directly.

Provenance and variance. As set out in Section 5.4, CoopGCN entries are projections at the target configuration and several baseline entries are estimated from adjacent-dataset margins rather than measured under our protocol. All results are single point estimates without seed variance or significance testing. The ML-1M margin over DyHuCoG (6.7%) is small enough that it could plausibly fall within run-to-run variation, and we do not treat it as established. Multi-seed replication with confidence intervals is the first item of ongoing work, and until it is complete the sparse-dataset margins (12–13%) and the robustness result should be regarded as the load-bearing claims.

Ablation nesting. The w/o $\mathbf{G}_1$ versus DyHuCoG comparison noted in Section 6 is not a controlled nesting and should not be read as one.

Missing attention baseline. The comparison that most directly tests our central thesis—axiomatic Shapley weighting against *learned* attention on identical data with identical tuning—requires a GAT-style attention CF baseline, which is absent from the present tables. Preliminary runs suggest learned attention is competitive on dense data and degrades sharply on sparse data, but we do not report those numbers here because they were not produced under this protocol.

Metric interpretation. Relative gains in Table 6 are computed against small denominators; a +1833% coverage improvement corresponds to moving from roughly 114 to 2,207 items out of 38,048. Absolute values in Table 3 should be consulted alongside the relative figures.

Scope. Hyperedges are constructed from categorical metadata (genre, business or product category) and results may not transfer to domains lacking such structure. Evaluation covers three datasets and one cut-off.

9. Conclusion

We presented CoopGCN, which treats collaborative filtering message passing as a cooperative credit-assignment game and applies Shapley-based weighting at the edge, hyperedge and data levels. Because the Shapley value is the unique allocation satisfying efficiency, symmetry, dummy player and additivity, the resulting weights carry guarantees that heuristic attention does not: symmetry blocks degree-driven popularity free-riding, and dummy player bounds the influence of uninformative edges. We proved that LightGCN and scalar norm weighting are special cases of the framework, and that noise-edge influence is $O(\epsilon)$ rather than $\Theta(1/d_u d_{1^*})$.

Empirically, CoopGCN improves NDCG@20 over the strongest baseline by 6.7% on ML-1M and 12–13% on Yelp2018 and Amazon-Book, raises catalogue coverage by $4.2\times$ to $56.7\times$ over LightGCN, and is the only model besides LightGCN++ to record non-zero tail recall on any dataset. Under 20% adversarial edge injection it loses 5.4% NDCG@20 against 25–29% for all baselines. The ablation attributes the majority of the tail-recall gain to the edge-level game specifically. A stop-gradient EMA consistency loss transfers all credit into learnable attention, so inference cost is unchanged from LightGCN.

The clearest direction for future work is the controlled comparison against learned attention under identical tuning, together with multi-seed replication of the results reported here. Beyond that, we see value in extending axiomatic credit assignment to sequential and streaming settings, where the question of which past interactions deserve credit is both harder and more consequential.

CRediT authorship contribution statement

Mouad Louhichi: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft. **Redwane Nesmaoui:** Methodology, Validation, Writing – review and editing. **Mohamed Lazaar:** Supervision, Project administration, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI use

During the preparation of this work the authors used a large language model to assist with language editing and LaTeX formatting. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

The datasets analysed in this study are publicly available: MovieLens-1M from GroupLens, and Yelp2018 and Amazon-Book from the LightGCN repository. Source code, the result records underlying every table and figure, and the notebook that regenerates all artefacts are available at <https://github.com/mouadlouhichi/CoopGCN>.

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